World-Model Value Depends on the Policy That Acts

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OUR POSITION

Evaluate world models together with the policies that act on their predictions.

A policy with better baseline performance can gain more from the same advice.

+11.50 mm

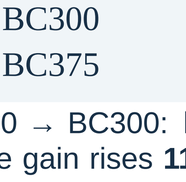
Change in advice gain
BC100 → BC300

01 Predict, score, select—then let the policy act.

Task: push an object to its goal (Meta-World / MuJoCo [1]). **BC:** behavior cloning from expert actions; policy input is simulator state.

Generate nine plans	→	Predict future features	→	Score each candidate	→	Keep or replace
1 BC plan + 8 modified plans 20 actions per plan		Frozen JEPA-WM [2] with a DINOv2 encoder		Learned distance estimate Lower score is better		Switch only if predicted gain exceeds a threshold

Forecast 20 steps; execute five actions. Scorers learn distance estimates from simulated outcomes.
WM scoring uses predicted features; current-feature scoring uses the same encoder without forecasting.



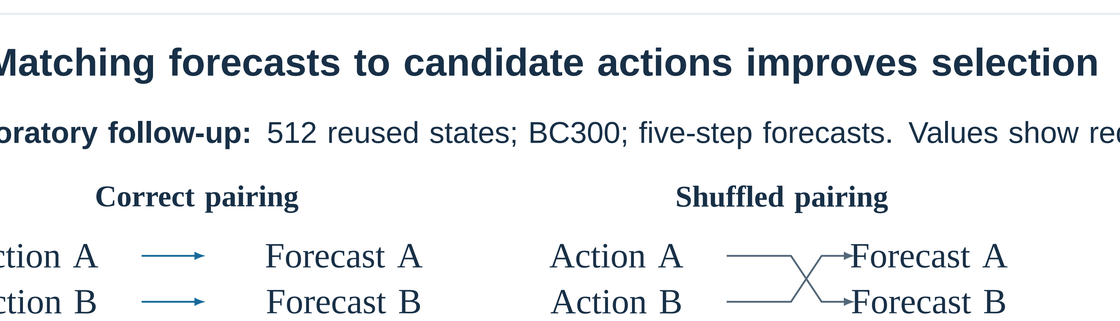
Identical starting state

Same saved state	Execute five stored actions	Same test policy in both runs	Final distance
	One initial intervention	Replan every five actions until step 100	Mean over steps 91–100
	Baseline BC300 actions	BC100, BC300, ...	Baseline error
	WM-selected actions	BC100, BC300, ...	Error with advice

Advice = five selected actions. Hold both action sequences fixed; vary only the policy used next. BC100/300: saved after 100/300 training updates.

02 Same advice, larger gain at BC300

512 shared states; one five-action intervention.
Scorer learns distance after 20 candidate actions.



Final distance and gain (mm)

Policy	Baseline ↓	With advice ↓	Gain ↑
BC100	143.51	146.64	**−3.13**
BC300	121.08	112.71	**+8.37**
BC375	115.36	105.77	**+9.60**

BC100 → BC300: baseline improves **22.43 mm**;
advice gain rises **11.50 mm** [6.80, 16.25].

Six checkpoints from one BC training run;
BC375 is a descriptive comparison.

03 Benefit changes across acting policies

1,024 cases; select repeatedly; baseline = BC alone.
Scorer learns final error after five actions + BC control.



World-model selection	Current-observation selection
+8.73 mm	**+18.04 mm**

At BC300, current-observation selection gains more.

WM gains are near zero at BC1000 / BC10000,
whose standalone performance is stronger.

BC300 success: **14.2% → 20.3% with WM (+6.2 pp)**.
Success: step-100 distance ≤50 mm; direct scoring: 35.4%.

Gain = baseline − advice error; positive is better. Bars and brackets: paired 95% intervals. Methods and adjusted tests: see paper.

04 Matching forecasts to candidate actions improves selection

Exploratory follow-up: 512 reused states; BC300; five-step forecasts. Values show reductions in final-task error.

Forecasts are shuffled across actions during training and evaluation.

Correct pairing	Shuffled pairing	Current features
17.08 mm	**5.48 mm**	**15.13 mm**

Matched − shuffled: **+11.60 mm** [7.37, 15.90].
Vs. current features: +1.95 mm [−0.96, 4.88]; advantage unresolved.

05 Robot examples: the same actions have different effects

Illustrative cases from the exploratory study. Images: step 100; distances: mean over the final 10 steps.

(a) Same advice, opposite effects on error

BC100		BC300	
Baseline	Advice	Baseline	Advice
9.6 mm	45.9 mm	82.4 mm	79.7 mm

Gain: −36.3 → +2.6 mm; baseline error increases.
Illustrative cases selected after evaluation.

(b) Both policies benefit, by different amounts

BC100		BC300	
Baseline	Advice	Baseline	Advice
243.3 mm	127.3 mm	243.3 mm	171.5 mm

Gain: +116.0 → +71.8 mm.

Evaluate world models with acting policies through task completion.

Evidence from one simulated task and one BC training run.

[1] Yu et al., *Meta-World*, CoRL 2020. [2] Terver et al., *What Drives Success in Physical Planning with Joint-Embedding Predictive World Models?*, 2025.

QR: Paper + robot videos